

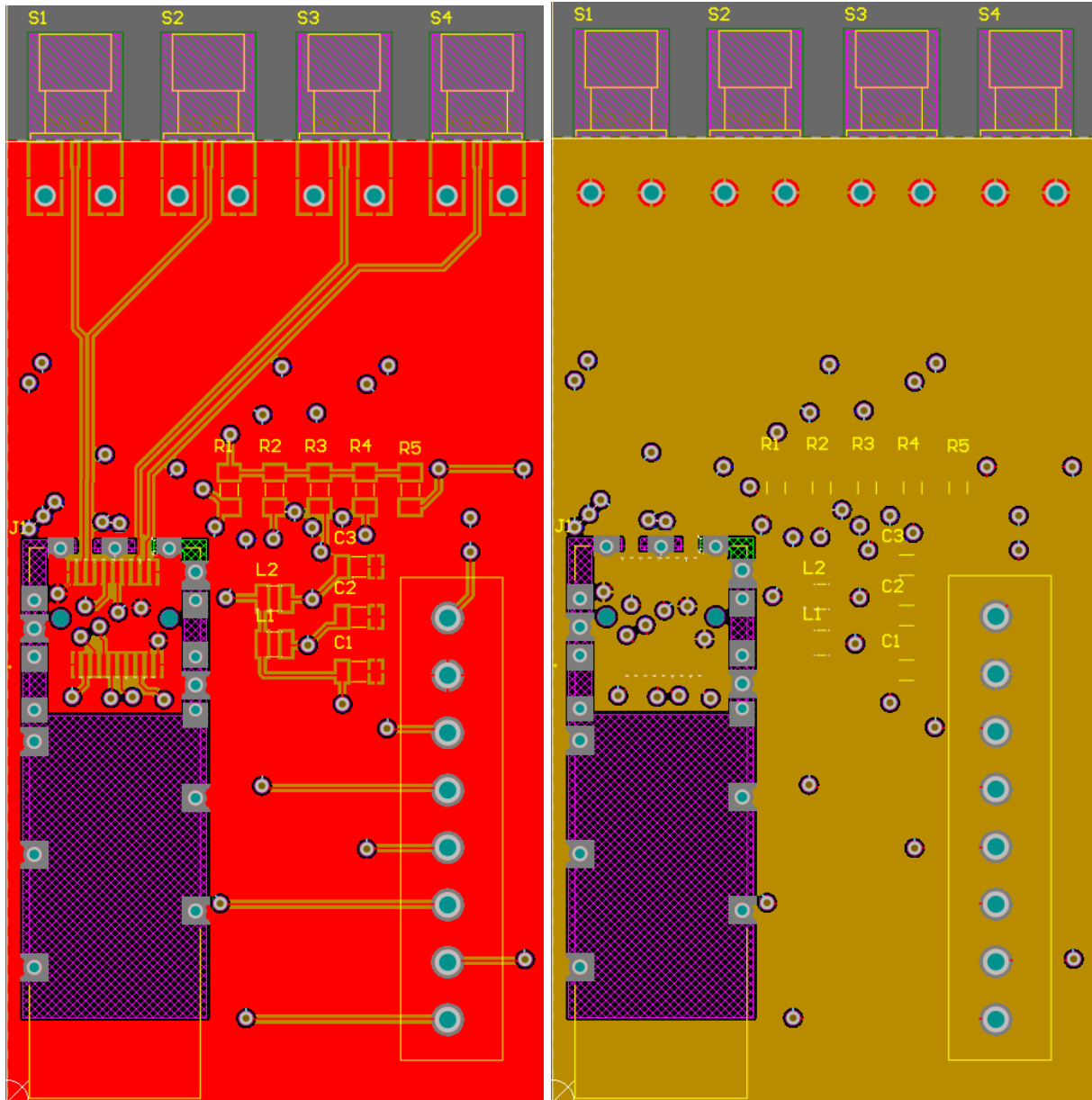


Figure 1: Top layer and Inner Layer 1

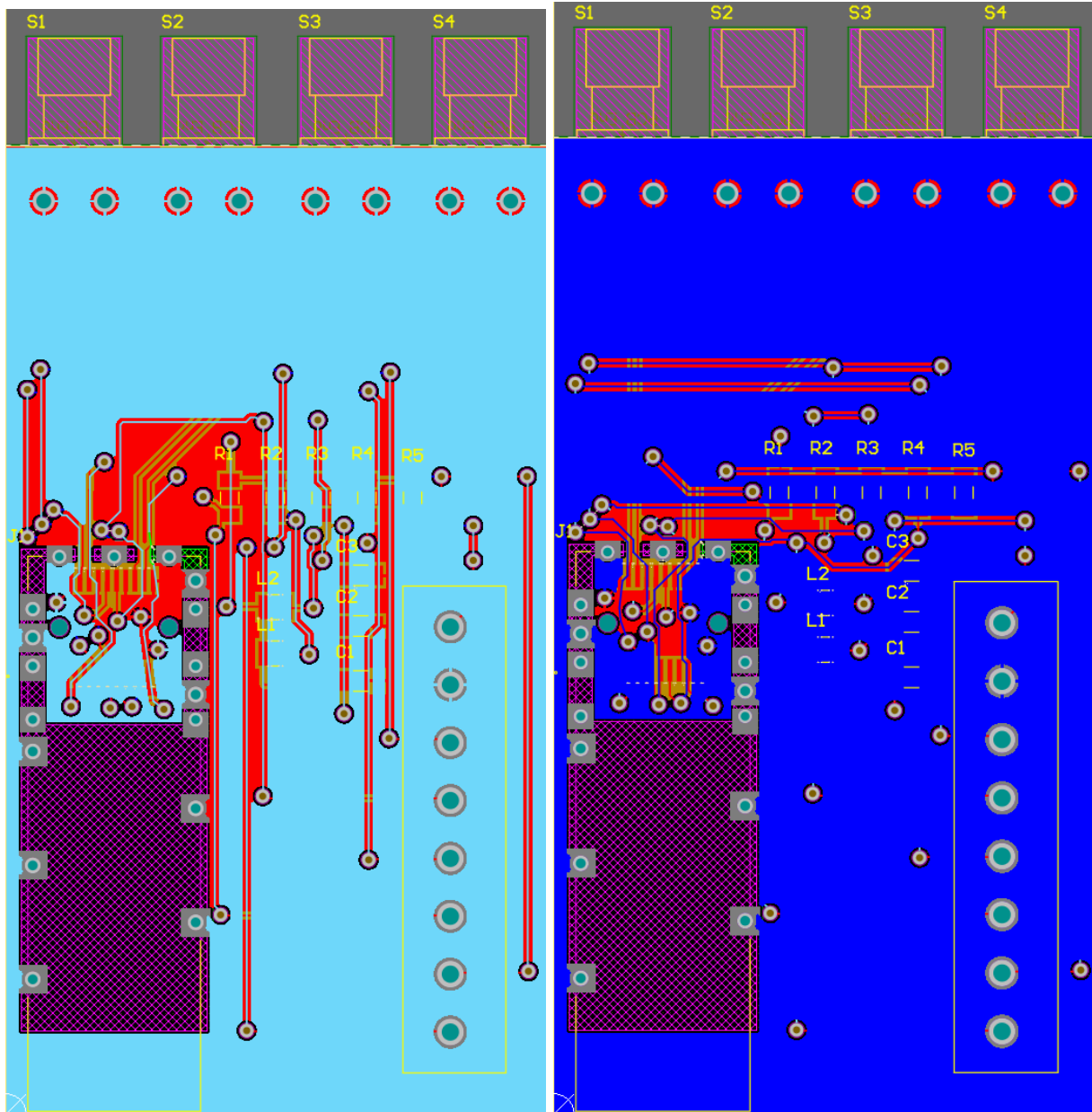


Figure 2: Inner layer 2 and Bottom layer

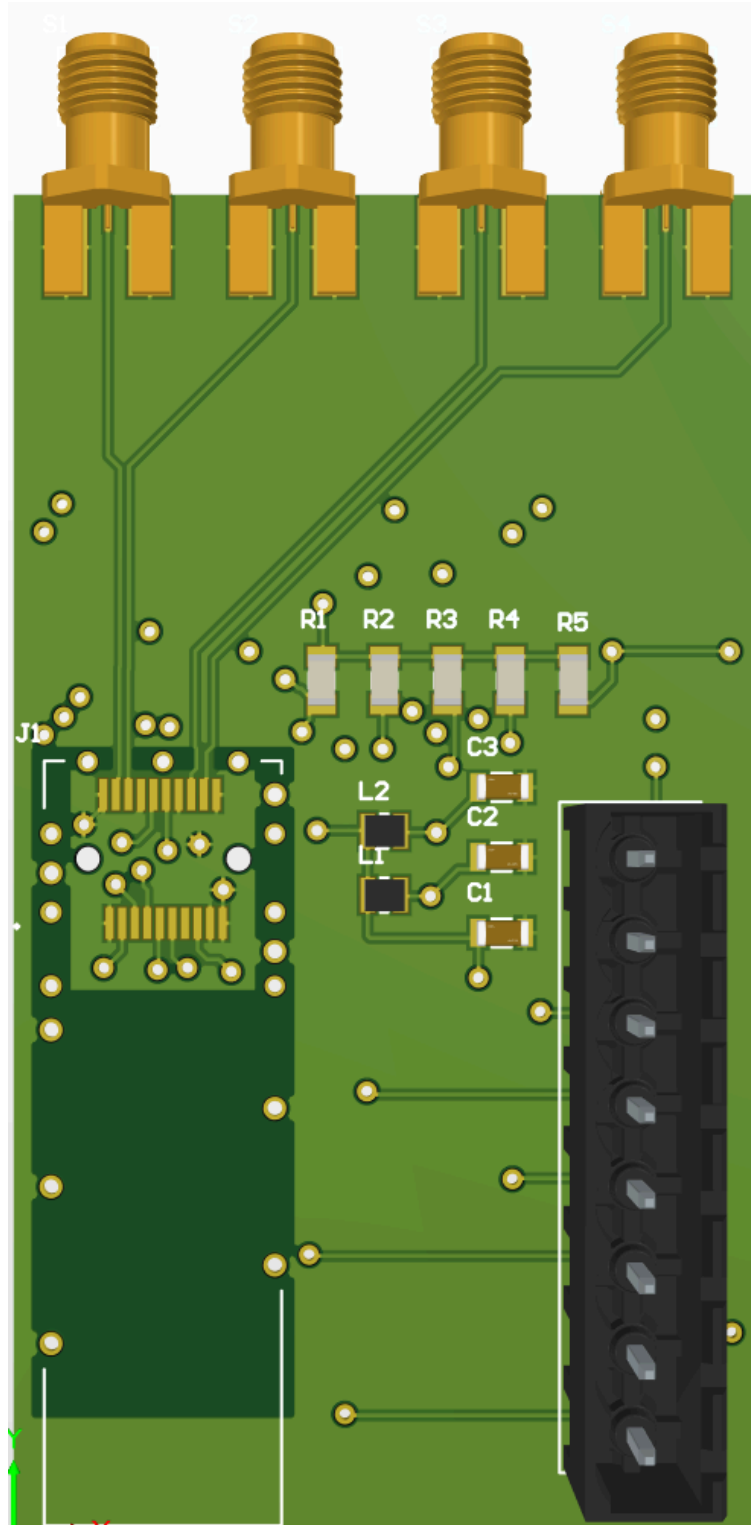


Figure 3: 3d representation

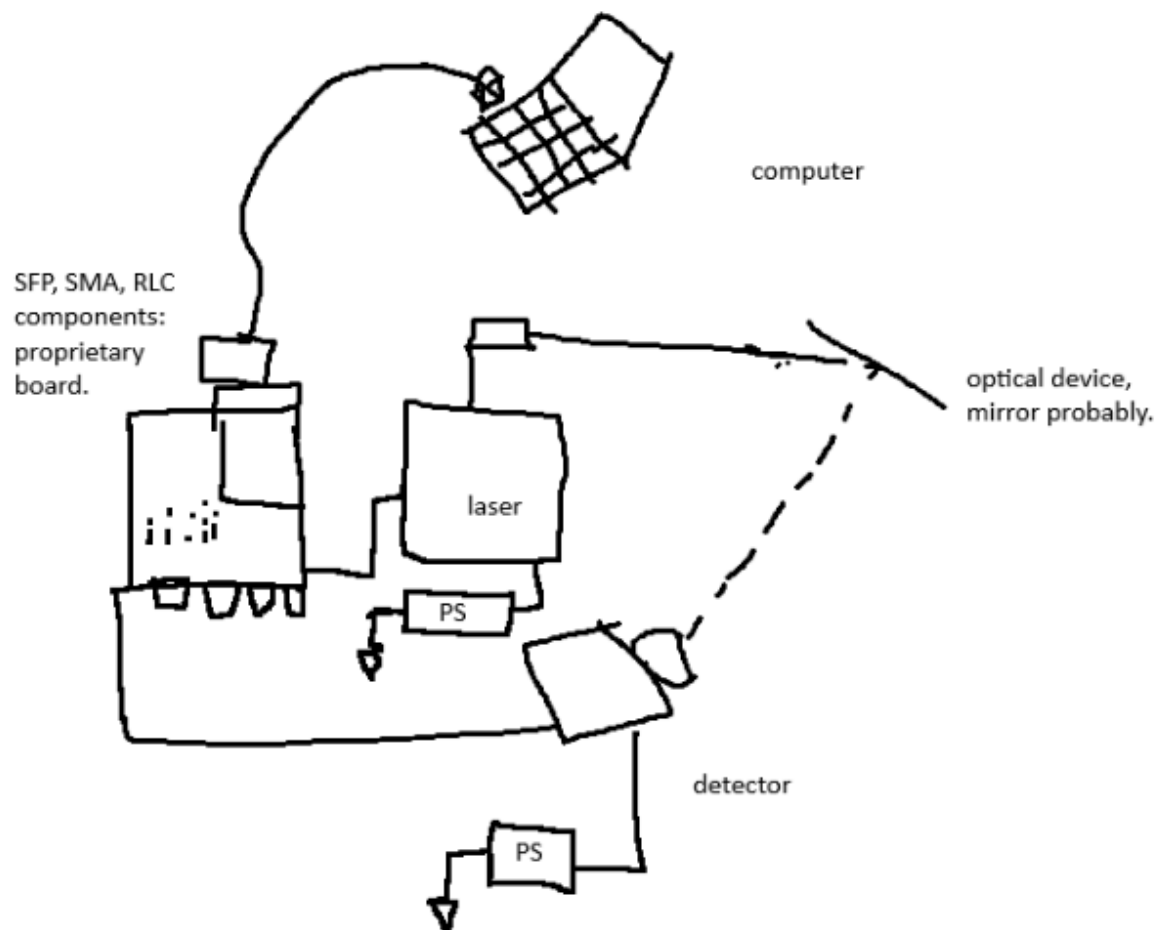


Figure 4: Proposed setup

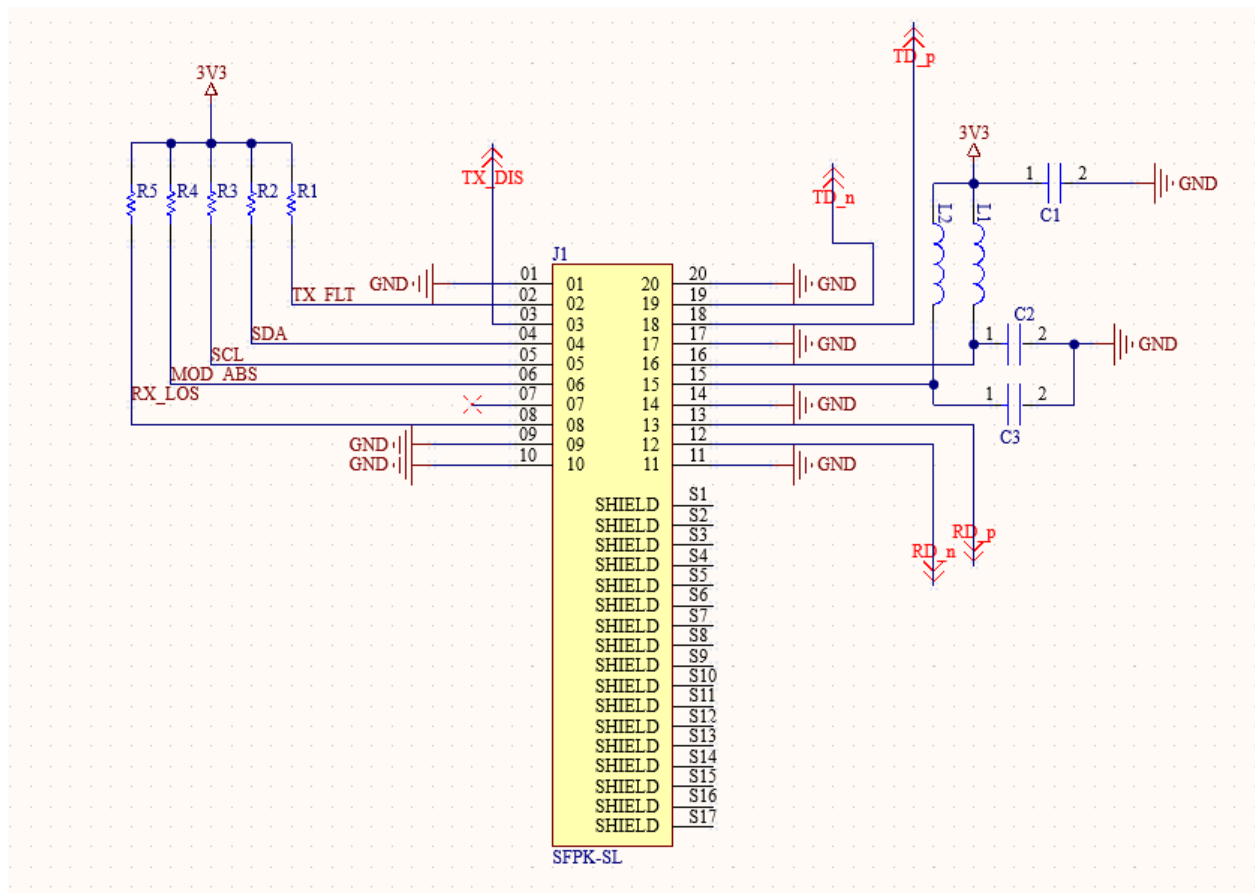


Figure 5: SFP schematic (Page 6 of provided data sheet)

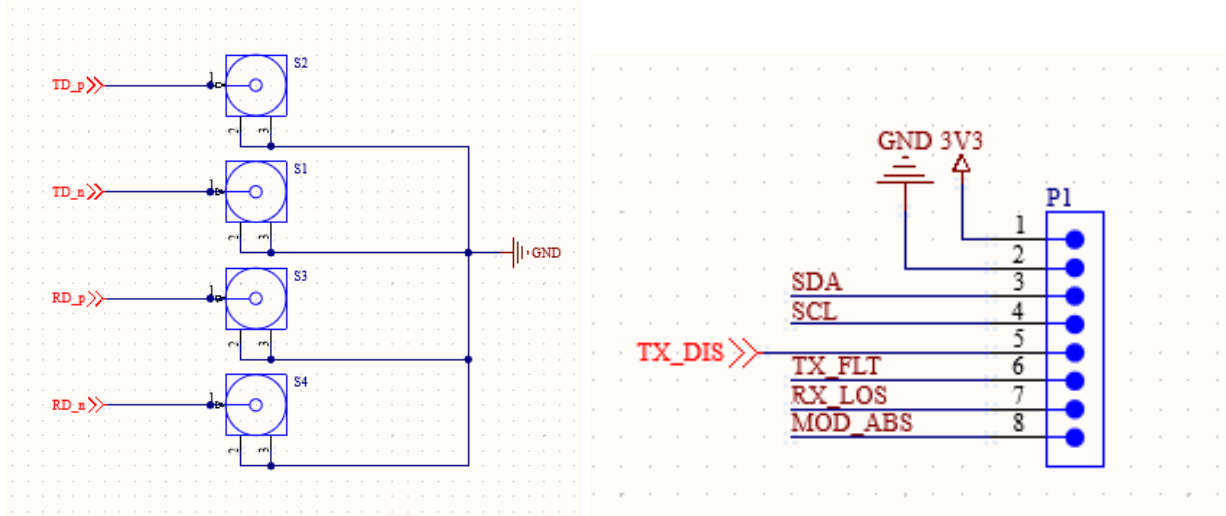


Figure 6: Header and Highspeed data schematics

This Mk1 design of the SFP board is routed to completion, and has only cosmetic errors. Things to improve for Mk2 could be differential pair nets distance difference shrinking, cosmetics, layout, and compactness. I had to switch to a 4-layer approach because the two slits at the top of the cage would only allow 2 nets to escape per layer ($2 * 2 * 2 = 8$), with 4 layers we can route 16 inner nets. From here, there are the following specification parameters to be clarified:

- Shall the line rate be 1 Gb/s or 10 Gb/s ?
- What laser and detector parts would be used in a hypothetical dream setup ?
- How does the laser interface with the SFP signals? differential, single-ended etc.
- If the PC is the host, would an FPGA or MCU be necessary?
- SMA vs GPPO for coax, I imagine SMA.
- Should CDR functionality be included?

That's about it.